TWO-COURSE PILOT | DISCUSSION DRAFT

Reliable computing for teaching

What we have previously relied on can no longer be guaranteed

Use ACCESS for the large CPU and GPU allocation and the classroom pilot. Use campus UITS/AWS only to test CPU components, and free GPU resources to benchmark workloads before piloting. Size the ACCESS request from measured demand; consider CloudBank through ACCESS only if its managed service can meet the confirmed requirements.

Planning baseline: 40 data science students + 40 MLIS students; six preparation weeks, 15 teaching weeks and two export/close weeks. Actual courses, term, instructors, PI, resource limits and operating commitments are pending.

Decision requested

Approve a preparation/canary ceiling of $8,000 within an illustrative total cash ceiling of $35,000. Release the remaining envelope only after the pre-class checks have dated evidence. Reserve committed faculty/Libraries/finance time separately. This is a proposed staged cap, not an existing authorization.

One component before a semester

Before piloting · preparation weeks 1–2 - Test components and benchmark GPUs

Exit: Dated CPU test evidence and GPU benchmark results, including failures and a justified workload envelope; no run is claimed here.

After benchmarks · before canary - Submit a sized ACCESS CPU + GPU request

Exit: Complete PI/course/term attachments and approved CPU/GPU resource configuration; request submission and award remain human/provider actions.

After award · preparation weeks 3–4 - Run a ten-person component canary

Exit: Both workflows succeed; actual service limits and unresolved defects are recorded. Stop or revise if core fit fails.

After canary · preparation weeks 5–6 - Rehearse the whole class and recovery

Exit: All thirteen applicable readiness checks have dated evidence and accountable sign-off; sponsor releases the remaining envelope.

Weeks 1–15 - Teach the two-course pilot

Exit: Record aggregate completion/access outcomes and actual costs; pause a failing component without removing the CPU course route.

Term end + two weeks - Export, close and decide

Exit: Verified exports, measured restore, closed accounts, final report and a costed continuation decision.

ASSIGNMENTS AND STAFFING

Two CPU workflows; named support

INFO 521: Introduction to Machine Learning - 40 planned seats

Build a reproducible data-cleaning and exploratory-analysis notebook, then fit a small CPU model with held-out validation. Explain missingness, leakage, uncertainty and data provenance.

Inputs: Generate a synthetic civic-service dataset with no real residents; keep core CSV inputs below 250 MB and test peak memory against the awarded workspace. Real/public datasets require a rights and data-classification check.

Environment: Requested: Python, JupyterLab, pandas, scikit-learn, matplotlib and an approved reproducible package manifest; R/RStudio only if the course requires it.

Acceptance: Run the whole assignment from a fresh student workspace; require execution without out-of-memory errors at the approved spec, reproducible key outputs and a complete exported submission.

LIS 671: Digital Curation and Digital Preservation - 40 planned seats

Audit a collection catalog, normalize names/dates/identifiers, document a metadata crosswalk, query an embedded catalog database and export a preservation-ready package with README, schema, provenance and checksums.

Inputs: Use a synthetic 10,000–100,000-record collection catalog with Dublin Core-style fields. Add public/licensed records only after rights review; exclude patron, circulation, donor and restricted collection records.

Environment: Requested: JupyterLab/Python, CSV/JSON/XML parsing and embedded SQLite; optional MARC parsing packages after approval. OpenRefine or other hosted applications require a separate provider fit check.

Acceptance: Execute the complete workflow in a fresh workspace; validate schema, record counts, identifiers and query results; reopen the exported package using the fallback environment.

Staffing coverage

Primary operator: four hours per teaching week, with 56 preparation and 12 close hours; backup: one hour/week, eight prep and four close. Two TAs: three hours per course/week plus four prep and two close hours each. Each course lead reserves 35 additional pilot hours. Normal course instruction/grading is outside this added-effort estimate.

Publish staffed help windows and a proposed one-business-day first-response target; no 24/7 promise. Escalation: TA/course lead to primary operator, backup/provider and service owner. Primary and backup operators must be different people.

GPU, MESA/CyVerse, NRP and OSPool modules need separate gates. PATh/IDSS eligibility is unconfirmed. Optional resources cannot become required assessment dependencies without their own checks.

ILLUSTRATIVE USD COSTS - NOT PAY SCALES OR QUOTES

Fund the operations, not only the resource

Role	Hours	Loaded rate	Mode	Cost
Service owner / allocation lead	24	$100	cash	$2,400
Primary technical operator	128	$90	cash	$11,520
Backup technical operator	27	$90	cash	$2,430
Two course support TAs	102	$35	cash	$3,570
Instructional design / accessibility	24	$75	cash	$1,800
IT security / privacy reviewer	12	$100	cash	$1,200
Two course leads	70	$90	in-kind	$6,300
Libraries / data steward	16	$85	in-kind	$1,360
College sponsor / finance	8	$70	in-kind	$560

Approved backup/export destinations: $2,000 reserve.

Durable exhibit / database destination: $600 reserve.

Fallback / unquoted service costs: $1,500 reserve.

Cash base $27,020 + 20% contingency $5,404 = $32,424. Required in-kind effort $8,220. Cash + committed-time value $40,644.

Illustrative USD planning rates, not UArizona pay scales, provider quotes or approved funds. Rates are assumed fully loaded (salary/fringe/overhead included); do not add fringe twice. In-kind effort is a required written workload commitment, not free labor. Replacing defaults changes the proposal only, not an approved budget.

ACCESS CPU/GPU access is covered only within an approved allocation and provider terms. Campus UITS/AWS is for component testing only; charges, duration and quotas are unquoted. Free GPU benchmarks have no planned paid compute charge but consume staff time and variable account quotas. CloudBank is conditional, with its own accounting. Zero additional provider charges in this calculator is a planning input, not a quote or award.

Excludes normal course instruction/grading, new staff recruitment, major platform development, laptops, paid software, optional GPU/model charges and institutional indirect costs beyond the assumed loaded rates. Additional scope requires a new estimate. No student-paid account or personal payment card is proposed.

Editable budget and conditional campus-storage comparison

THIRTEEN CHECKS - ALL PLANNED / NOT RUN

Verify before class.

1. Scope | Sponsor + service owner

Named eligible PI, leads, distinct operators and funded term coverage.

2. Benchmark | Primary operator + course leads

Campus AWS CPU tests and free GPU benchmarks completed; workload manifest, failures and target ACCESS replay recorded.

3. Provider | Allocation lead + selected ACCESS provider

ACCESS CPU/GPU exchanges, actual quotas/storage, accounting and educational project dates confirmed; CloudBank conditional.

4. Accounts | Operator + course leads

Approved logins, isolation and account revocation demonstrated.

5. Software | Each course lead

Complete assignments run in the pinned environment; required GPU work fits the verified ACCESS device.

6. Capacity | Operator + provider

Provider-approved class load rehearsal; proposed p95 workspace access within five minutes.

7. Save | Course leads + TAs

Save/restart/export demonstrated and outputs reopen elsewhere.

8. Grades | Course leads

Versioned submissions and test grade receipts reconcile in the approved LMS.

9. Data | IT/privacy reviewer + Libraries

Data rights, access roles, destinations and retention approved.

10. Accessibility | Instructional designer + course leads

Essential keyboard/screen-reader workflow and equivalent assessed CPU route demonstrated.

11. Recovery | Primary + backup operators

Independent restore measured against proposed 24-hour RPO and one-business-day RTO.

12. Support | Owner + TAs + operators

Staffed support, escalation and outage fallback rehearsed; no 24/7 promise.

13. Exit | Service owner + course leads + Libraries

Expiry warnings, learner/grade exports, closure/deletion and next-term decision documented.

Evidence, evaluation and exit

Keep private account/grade evidence in the institutional register. Review aggregate access, assignment success, exports, support hours, costs and restore timing. Reconcile grades and complete export, retention and closure before expiry.

Full readiness register and evidence criteria